Republic Of

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Department of Computer
Faculty of Engineering and
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Ministry of Higher Education



VR BRIDGE

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A graduation project report presented to the Computer Science Department in partial fulfillment of the requirements for a Bachelor's degree in Information Technology.

A Bridge Between the Human and his Future

By

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Pledge

We, Khalil ALqisy and Mohamed Qasem, hereby pledge my commitment to the successful completion of my graduation project, titled VR Bridge. I solemnly affirm that all work presented in this project is the result of our own efforts, and I have upheld the highest standards of academic integrity throughout the research and development process.

We pledge to:

1. Conduct thorough and ethical research.
2. Accurately represent all data and findings.
3. Properly cite all sources and references used in our work.
4. Seek guidance and support from our mentors and peers as needed.
5. Maintain open and honest communication with our project advisors.
6. Continuously strive for excellence and rigor in our work.

We acknowledge that this project is a significant component of our academic journey, and we are committed to dedicating the necessary time, effort, and resources to ensure its successful completion. By signing this pledge, I affirm my dedication to upholding the principles of honesty, responsibility, and academic excellence.

_____

Khalil Alqisy

Moahmed Qasem

2023/2024

Acknowledgments

I would like to express my deepest gratitude to those who have supported me throughout the completion of my graduation project.

First and foremost, to my parents: your unwavering support, encouragement, and love have been the foundation of my academic journey. Your sacrifices and belief in me have made this accomplishment possible.

To my siblings: your constant motivation and understanding have been invaluable. You have been my cheerleaders, and I am grateful for your patience and support.

To my friends: thank you for standing by me through the ups and downs, for your encouragement, and for providing the much-needed balance between work and leisure. Your friendship has been a source of strength and joy.

To my supervisor, Dr. Mohanad Almoshriqy: your guidance, insights, and mentorship have been instrumental in the successful completion of this project. Your expertise and dedication have greatly enriched my learning experience, and I am profoundly thankful for your support.

This project is dedicated to all of you as a token of my appreciation and gratitude. Without your support, this achievement would not have been possible.

Thank you.

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Phase One

INTRO

INTRODUCTION

Today’s virtual reality technologies build upon ideas that date back to the 1800s, almost to the very beginning of practical photography. In 1838, the first stereoscope was invented, using twin mirrors to project a single image. That eventually developed into the View-Master, patented in 1939 and still produced today.

The use of the term “virtual reality,” however, was first used in the mid-1980s when Jaron Lanier, founder of VPL Research, began to develop the gear, including goggles and gloves, needed to experience what he called “virtual reality.”

Even before that, however, technologists were developing simulated environments. One milestone was the Sensorama in 1956. Morton Heilig’s background was in the Hollywood motion picture industry. He wanted to see how people could feel like they were “in” the movie. The Sensorama experience simulated a real city environment, which you “rode” through on a motorcycle. Multisensory stimulation let you see the road, hear the engine, feel the vibration, and smell the motor’s exhaust in the designed “world.”

Heilig also patented a head-mounted display device, called the Telesphere Mask, in 1960. Many inventors would build upon his foundational work.

By 1965, another inventor, Ivan Sutherland, offered “the Ultimate Display,” a head-mounted device that he suggested would serve as a “window into a virtual world.”

The 1970s and 1980s were a heady time in the field. Optical advances ran parallel to projects that worked on haptic devices and other instruments that would allow you to move around in the virtual space. At NASA Ames Research Center in the mid-1980s, for example, the Virtual Interface Environment Workstation (VIEW) system combined a head-mounted device with gloves to enable the haptic interaction.

Today’s current virtual reality gear owes a debt of gratitude to the pioneering inventors of the past six decades who paved the way for the low-cost, high-quality devices which are easily accessible.

Be sure to visit the VR flight simulators at The Franklin Institute to experience a virtual environment yourself![1]

Now a days When you say VR the first thing That came to minds is Headset with Lenses, and Controllers.

Headsets Used to use External camera system to track the position of both the headset and controllers, but now Companies like Meta (Oculus previously) making portable headsets, thanks to it’s built in 6DoF camera that can do the same job as external ones.

In the other the hand, controllers are tools to translate your hands position and rotation, it’s also Contain some buttons to do some Functions Such as grab or select, new controllers like the ones in Meta Ques Pro have their own cameras and can track themselves.

But is that enough?

Let’s move on to See!

SUMMARY

Virtual Reality Bridge (VRB) is a glove which can track your hand and fingers movement and monitor it in VR, and do feedback based on the objects that you hold/touch or interact with, this can be helpful in training in multiple majors, such as medical, military, instructor, learning, etc…...

PROBLEMS

The VR industry is still in his begging and in early stages, and need a lot of work to be in form for all users.

And the sub problems are showed below:

1- Actually, to now there's no full hand tracking or force feedback for real or at least as a product.
2- The VR haptics are very expensive.
3- The lack of emulators that can make learning and other industries easier.

GOALS

Make VR more like a tool that can make our everyday life easier.

And the goals for sub problems is showed below:

1- Make VR services available for all.
2- Build VR learning & entertaining.
3- Make learning easier and more affordable.

Phase Tow

THE PRIMARILY STUDY

LITERATURE REVIEW

I did an online search and found the following Companies related to My project.

1- **Meta (Oculus Previously)**

In 2012, a new Kickstarter campaign took the internet by storm. Palmer Luckey introduced the concept and the first test series for the Oculus Rift and Oculus Touch controllers. The fundraising campaign raised $2.4 million, tenfold of the $250,000 the campaign aimed to achieve. The funds allowed Oculus to push out two pre-production models for developers to work with.

Before the hardware had come into fruition, Facebook, Inc saw the potential for what VR can be. In March of 2014, Facebook acquired Oculus for $2.3 billion in both cash and stock. This move proved to be beneficial in the long run for Virtual Reality headsets. Now a part of Facebook, Inc, Oculus moved its headquarters to Menlo Park.[2]



Figure 1 – all Oculus VR Headsets ever released

2- Valve



Figure 2 – Valve Index VR headsets

3- HTC



Figure 3 – HTC VIVE VR headsets Lineup

4- HP

It’s Another Company that Make VR Headsets and Tools.

Phase Three

METHODOLOGY AND PPLANNING

METHODOLOGY

1.Design

The Design Phase will include the following aspects:

1.1 Analyze Required Parts

Which parts are Required to accomplish the full design for the Haptics and battery pack etc...

1.2 3D Design

All parts will be designed in blender and tested using emulators.

1.3 Design the User interface

Design VR Demo for testing.

2 Implementation

The Implementation Phase will include the following aspects:

2.1 Printing

All parts That can’t be found in the market will be 3D printed using ender 3D printer.

2.2 install parts together

I will install all parts and plug power adapter and ensure everything work fine.

2.3 Develop the test demo in Engine

Using unity 3D or Unreal Engine the demo will be built

2.4 Make Steam Drivers

Based on the Arduino config, Drivers Will be built to be recognized.

3 Testing

During the development process, unit testing will be done to ensure all modules are built correctly. System integration testing will be done after I have built all the components and combined them into the Devices. I will test the Demo, the Drivers and the Accuracy.

3.1 Test the Main devices

To test VR haptics, Battery pack….

3.2 Test the Demo

To test the Demo and its interaction with VR haptics.

3.3 Test the Drivers

To Ensure all Movements are translated between the haptics and PC

3.4 Test Accuracy

To Ensure the that everything installed right and work as intended.

4 Evaluation

At the Ending I would like to say, these opinions are just suggested by me and doesn’t have to be right and I came up with all of this as VR user and fan for long time.

I hope someone get inspired or at least get some knowledge about VR and what it has to offer for you.

PROJECT PLANNING

1-Tasks

TASK
Do the Literature Survey
Analyze required parts
Design 3D parts
Design prototypes
Design Demo
Print 3D Parts
Build Parts together
Build demo in Engine
Establish Steam drivers (Open VR)
Test Devices and Printed Parts
Test Demo app
Test the Drivers
Test the Accuracy
Write the Progress Report
Write the Final Report

2- GANTT Chart

TASK	Apr	Apr	Apr	May	May	May	Jun	Jun	Jun	July
Do the Literature Survey										
Analyze required parts										
Design 3D parts										
Design prototypes										
Design Demo										
Print 3D Parts										
Build Parts together										
Build demo in Engine										
Establish Steam drivers (Open VR)										
Test Devices and Printed Parts										
Test Demo app										
Test the Drivers										
Test the Accuracy										
Write the Progress Report										
Write the Final Report										

Phase Four

HARDWARE AND SOFTWARE

HARDWARE, SSFTWARE REQOUERED AND COST

1- Hardware

A- Development PCs:

1- Lenovo legion 7i with i7 11th gen, RTX 3070 and 16gb of ram.

2- Asus rog Zephyrus duo 15 with i9 10th gen, RTX 2080 super and 32gb of ram.

3- Asus TUF dash 15 with i7 11th gen, RTX 3070 and 40gb of ram.

B- Derives of Design and testing:

4- Meta quest 2 VR headset. 400$

5- Creativity CR-M4 3D printer. 1000$+

6- ESP-WROOM-32 (1x per hand). 11$

C- Glove Parts:

1- b10k ohm Potentiometer (5x per hand) 2.00$

2- 9g servo motors for haptics(5x per hand) 20$

3- retractable badge reels (for the rotary springs, 5x per hand) 4.00$

4- Glove material 1.00$

5- Mounting material 0.50$

6- Breadboards tools for using JST connectors, 5 pin JST-XH 2.54mm 5-10$

7- source of power (5v)

Hardware Cost:

COMPONENT	QTY	PRICE	TOTAL
LENOVO LEGION 7 GEN 6	**1**	**1600$**	**1600$**
ASUS ROG ZYPHROUSE DOU 15	**1**	**1800$**	**1800$**
ASUS TUF DASH 15	**1**	**1400$**	**1400$**
META QUEST 2	**1**	**400$**	**400$**
CREATIVTY CR-M4	**1**	**1000$**	**1000$**
ESP323	**2**	**11$**	**22$**
B10K POTS	**20**	**0.40$**	**8$**
SERVO MOTORS	**5**	**4$**	**20$**
3D FILAMENT	**1**	**25$**	**25$**

D- 3D Printed Parts:

1- HapticSpool (5x)
2- Tensioner_GreenPot (5x) OR Tensioner_WLPot (5x)
3- SpoolCover (5x)
4- RigidMount (1x) (Left or right hand)
5- WLPotSlideHolder OR WLPotElasticHolder OR GreenPotSlideHolder (5x)
6- Quest2_MountSlider OR Vive3.0_MountSlider (1x)
7- GuideRing (Will need to be resized) OR GuideNode (2+ per finger, 1+ for thumb)
8- EndCap (1x per finger, will need to be resized to fit)

3D Filament cost 30$

2- Software:

1- Blender For 3D design
2- Cura Slicer for 3D printer Gcode Generating.
3- Oculus and Steam VR for drivers and Handling.
4- Unity 3D and Unreal Engine for demo development.
5- Arduino Software for Coding.

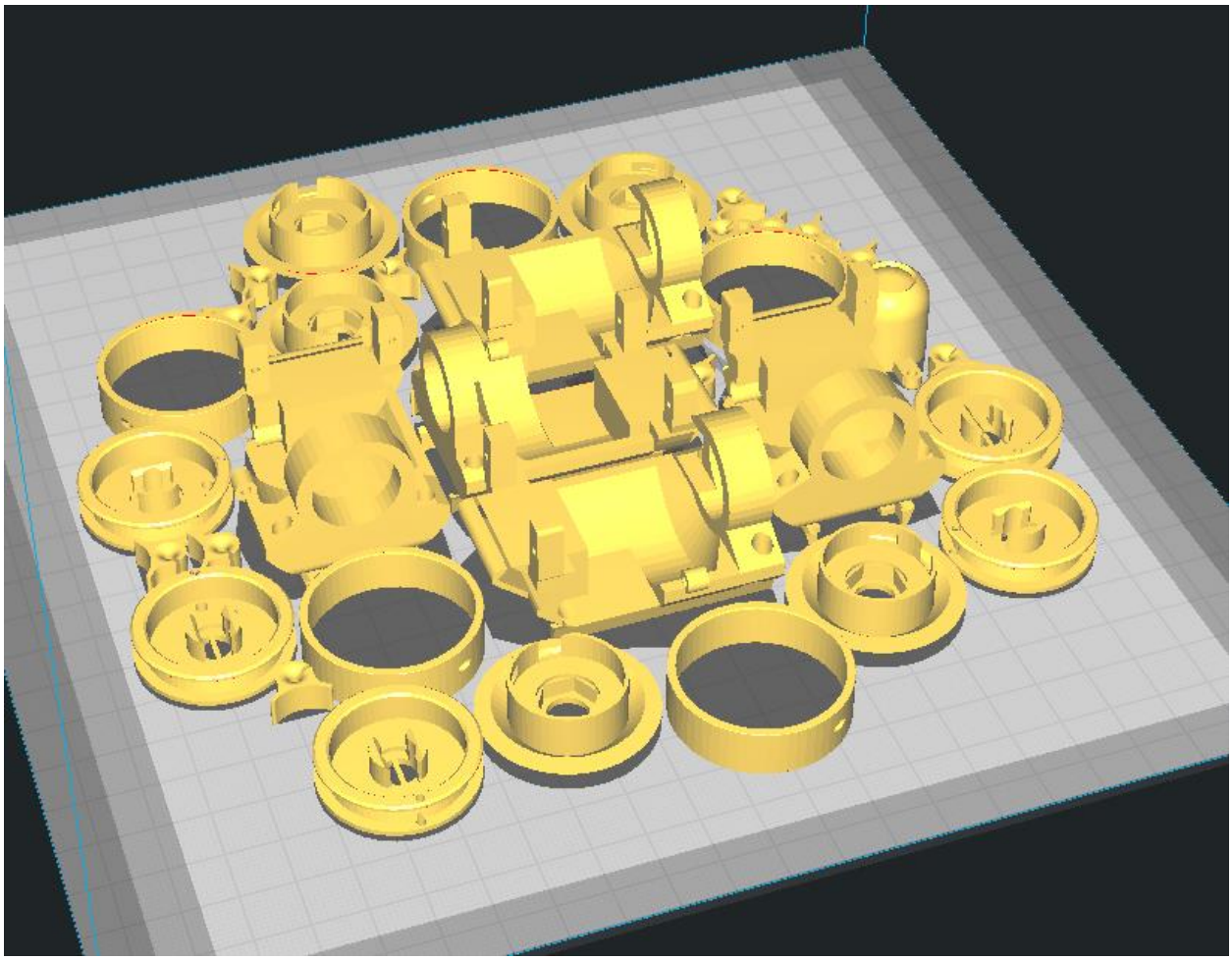
Software Cost:

COMPONENT	QTY	PRICE	TOTAL
ARDUINO IDE	**3**	**Free**	**Free**
WINDOWS 11 PRO	**3**	**20$**	**60$**
MICROSOFT OFFICE 2021	**3**	**30$**	**90$**
META LINK	**3**	**Free**	**Free**
STEAM VR	**3**	**Free**	**Free**
CURA SLICER	**3**	**Free**	**Free**
BLENDER DONATION	**1**	**20$**	**20$**
UNITY 3D	**1**	**Free**	**Free**
INTERNET FEES	**2**	**30$**	**60$**

Phase Five

DESIGNING AND WIRING

1- Assembly



1- First, we used the POT and plug it to the ESP32 to read the current voltage and make sure it does the effect needed to achieve the desired movement.

2- Then use WD40 in the POT crack to make them loose, so it can be retracted using a coil, and make easy of movement.

3- The coil will be installed inside the tensor (see the parts list) and attached to the POT, and then covered with the haptic spool.

4- We put all of that inside the mounter, and then installed the servo and screws.

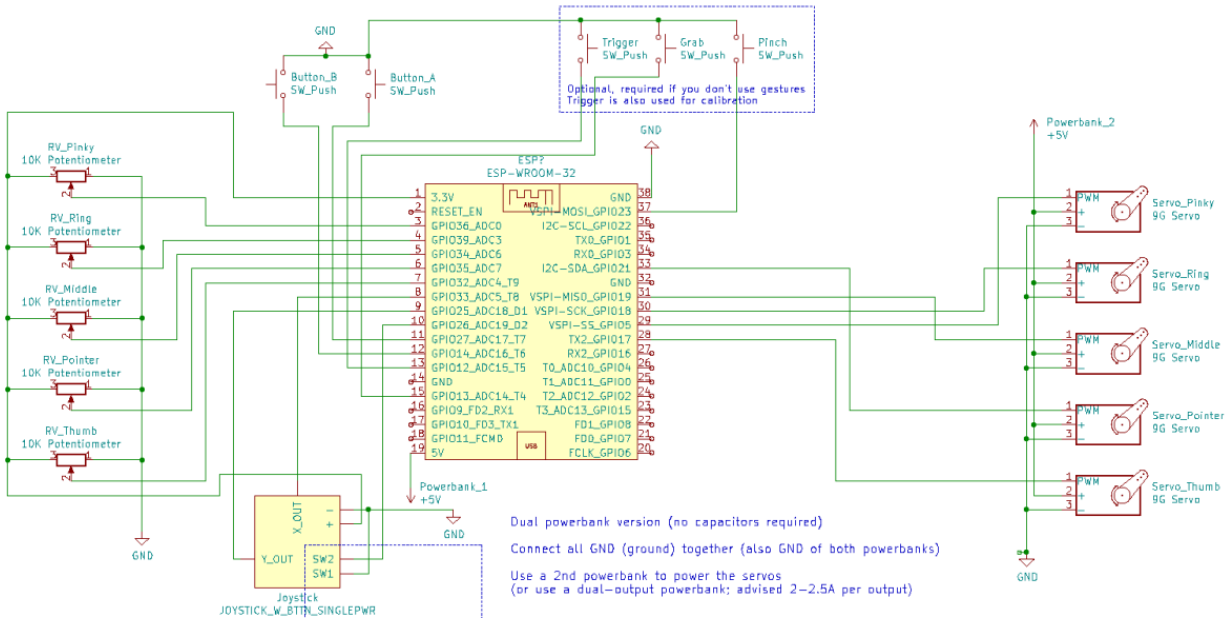
5- The whole unit will take place upon the slider along the quest mounter for position tracking.

2- Wiring

1- Every POT and Every Servo have three wires One VCC, One GND, and One Signal that go directly to the board.

2- For the POTs 3.3 pin and GND was used with 5 of analog Pins.

3- For the Servos External Power Source was used with 5V and 5 Digital Pins.

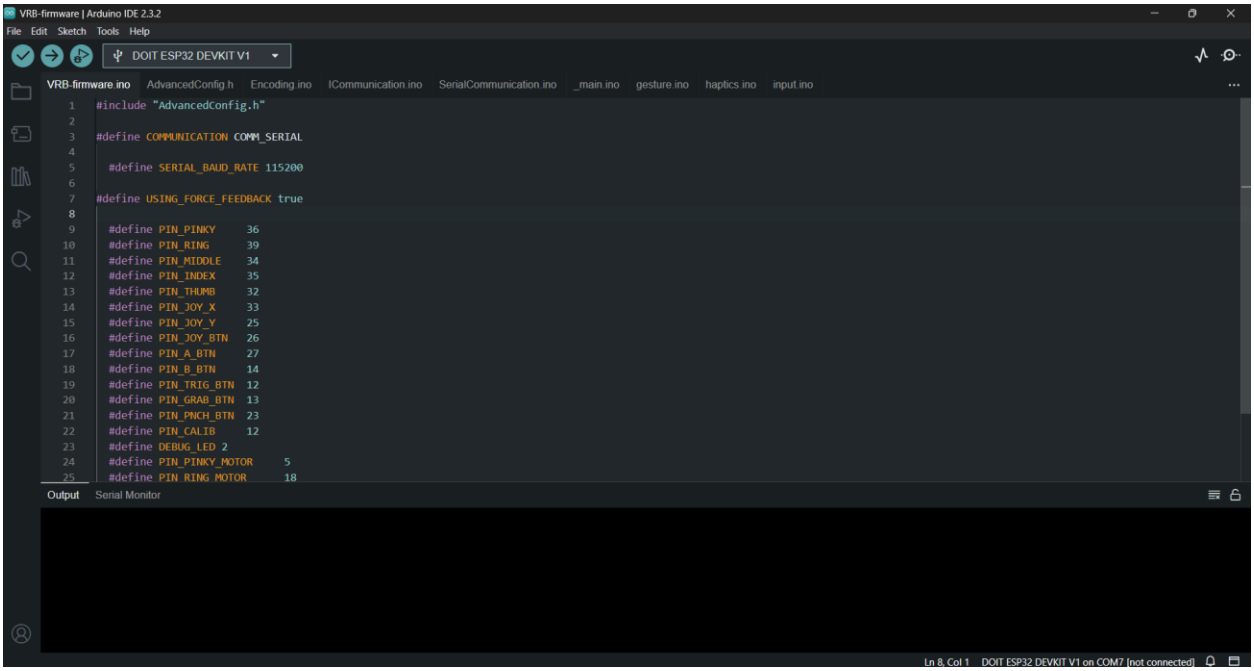


Input	*Pin*
Pinky Finger	36
Ring Finger	39
Middle Finger	34
Pointer Finger	35
Thumb	32
Pinky Motor	5
Ring Motor	18
Middle Motor	19
Pointer Motor	21
Thumb Motor	17

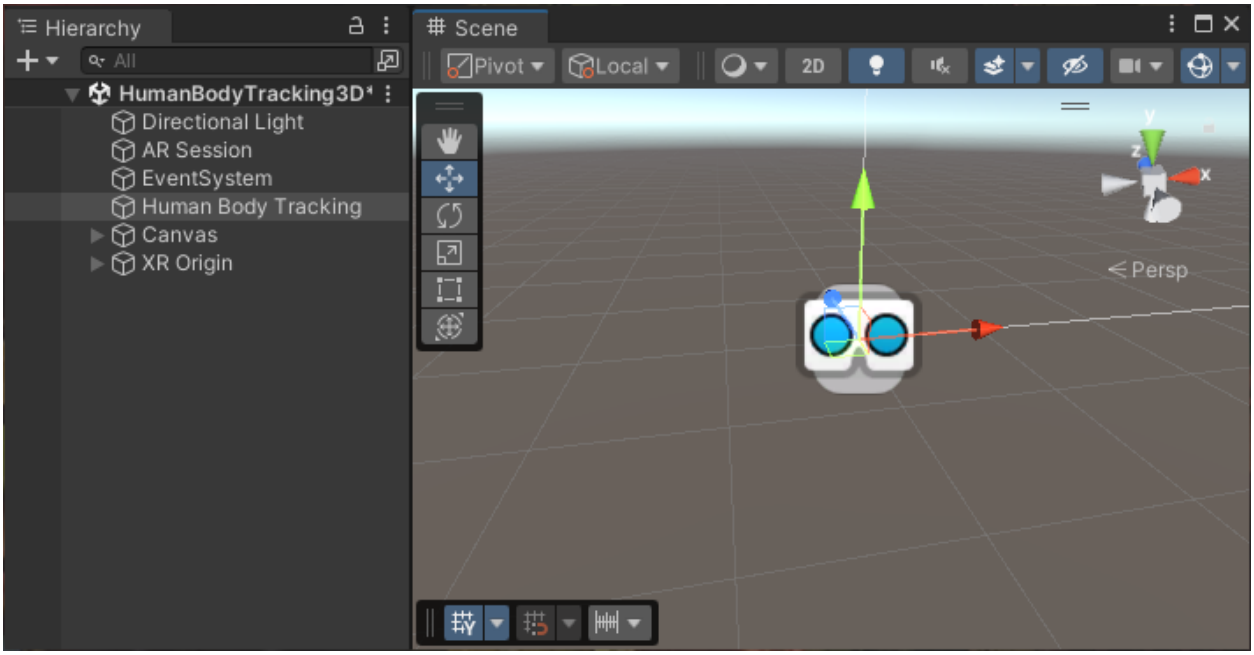
3- Coding and Demoing

The code main function is to translate POT reads to values used by open gloves drivers to emulate finger movement, and to send signals to servos in order to stop the spool in the desired angle based on the object collision in demo, which can make a sense of felling of motion limitation**.**

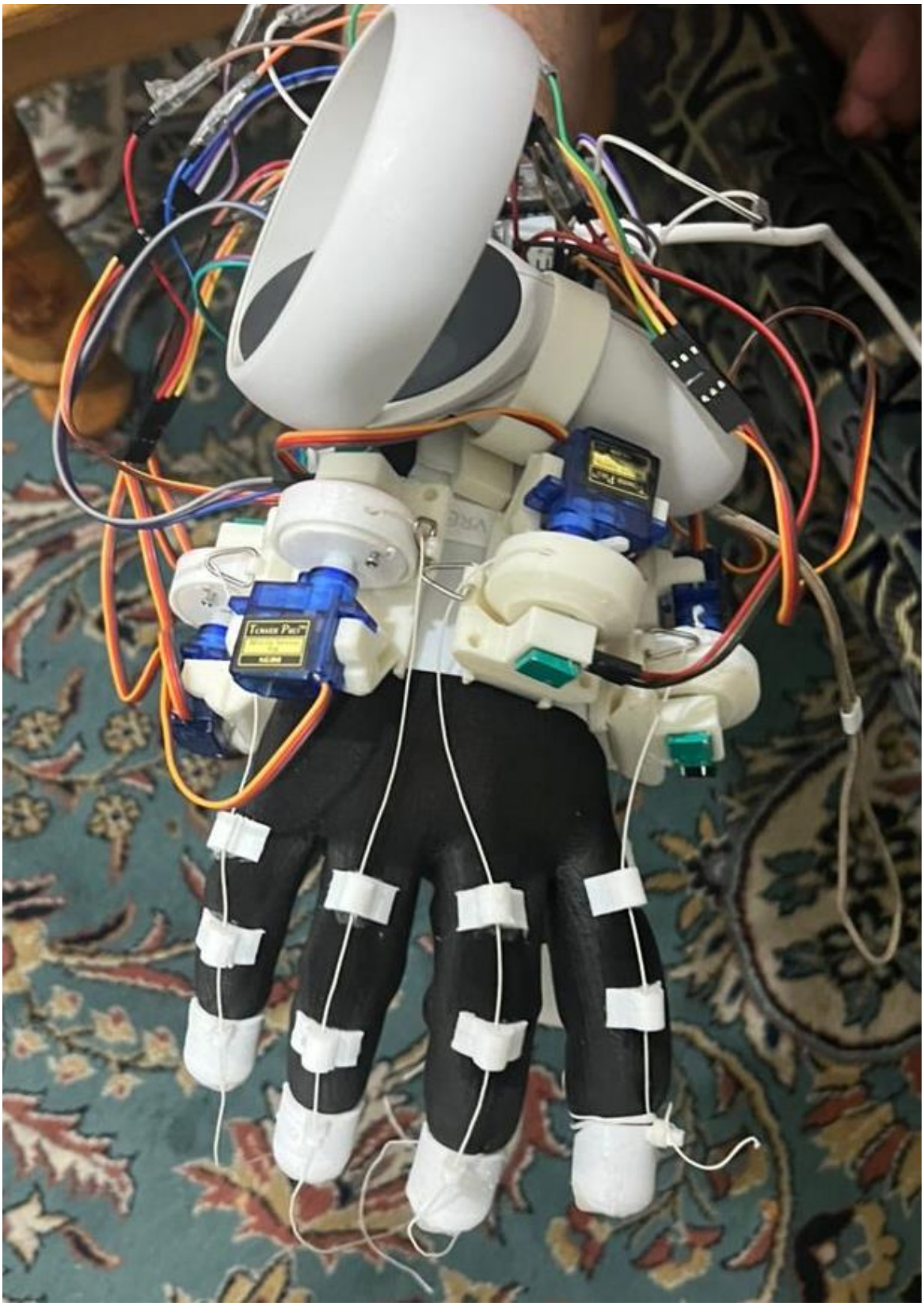
For the demo it uses Unity 3D and it paired with the drivers to establish connection wired or over Bluetooth.



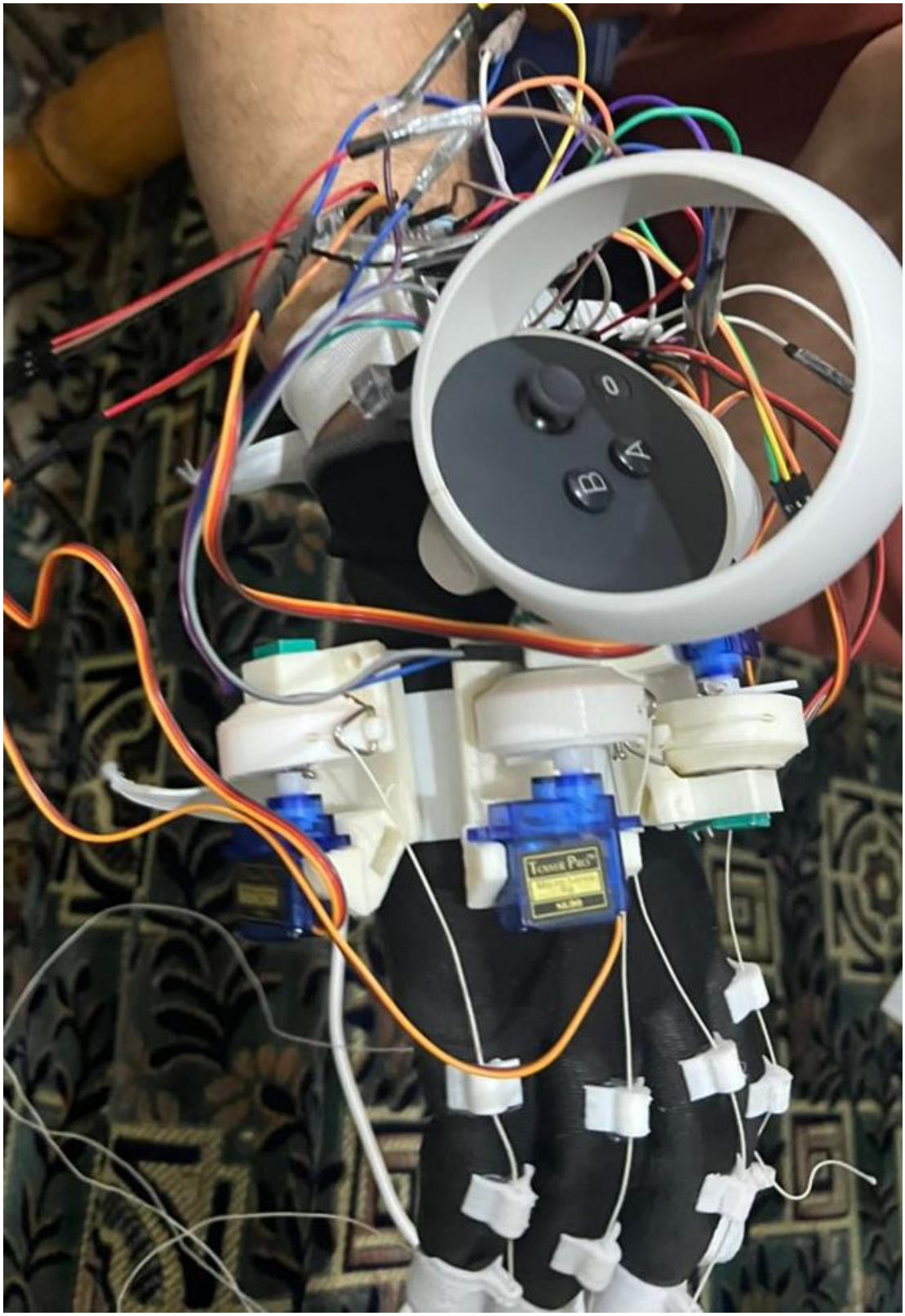
Arduino IDE



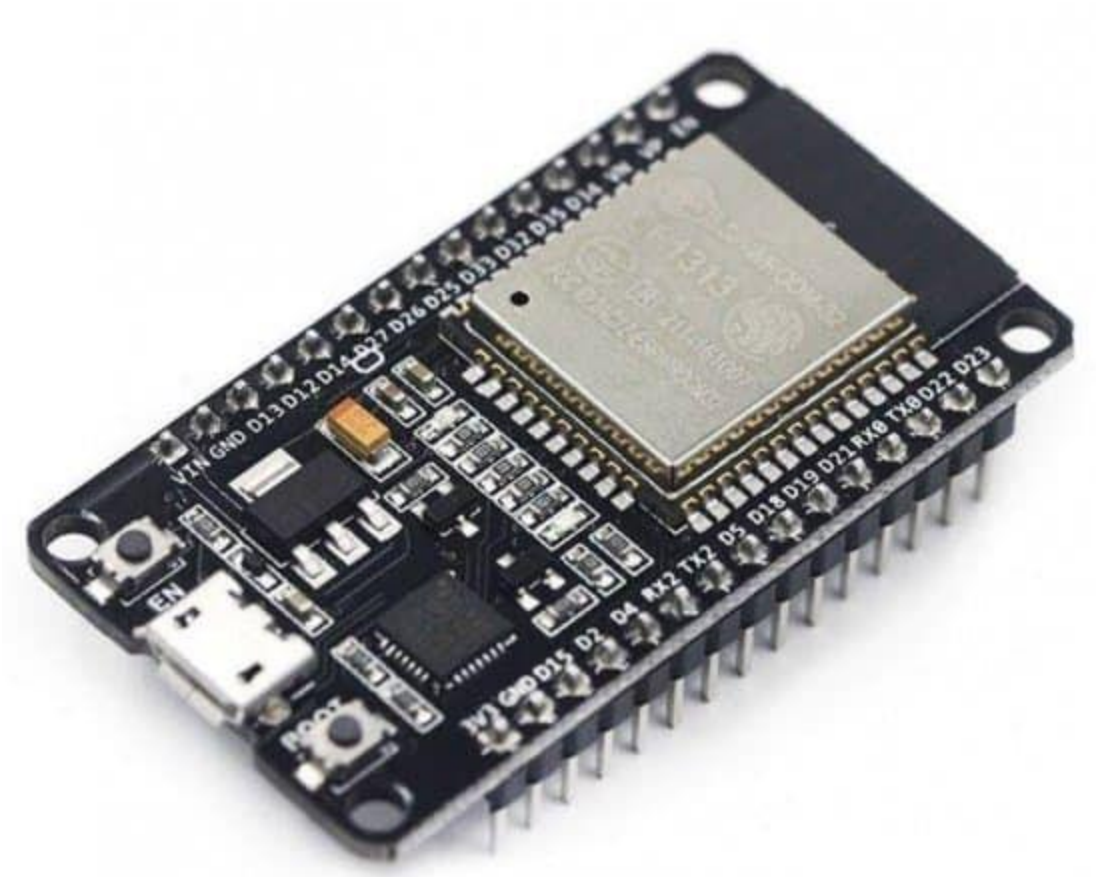
Unity 3D



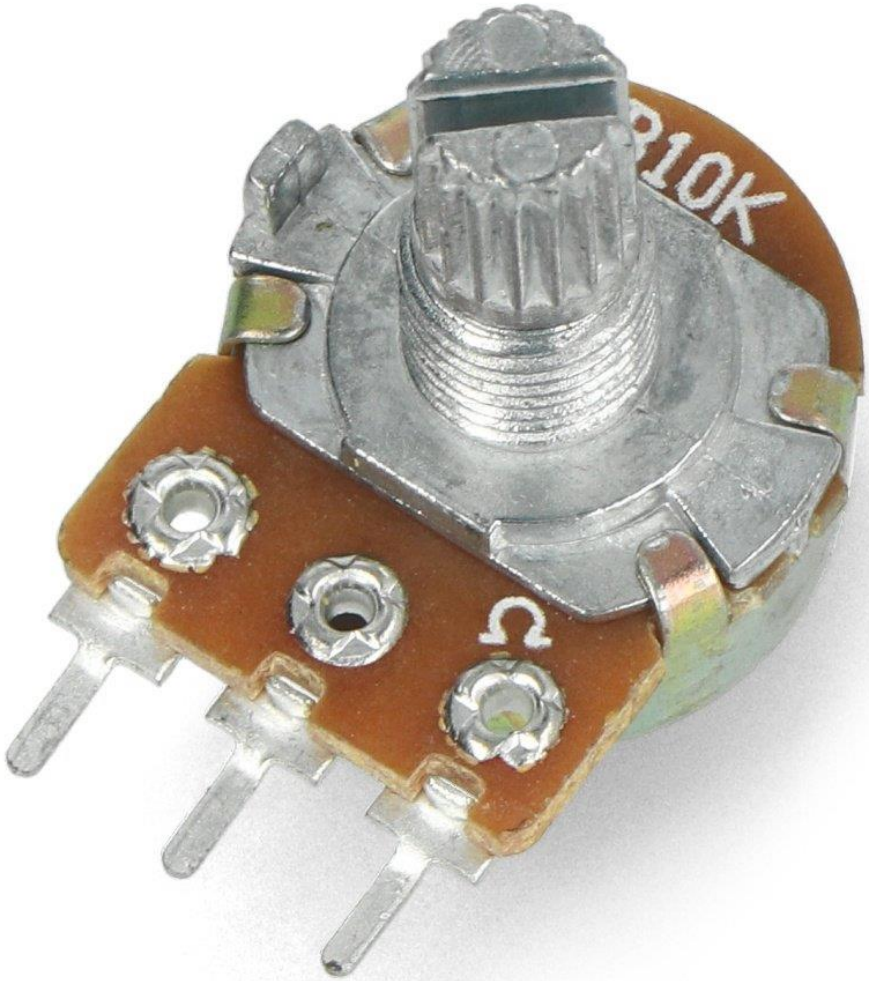
VRB (1)



VRB (2)



ESP32 Dev Kit (3)



B10K WL Potentiometer (4)



Servo Motor (5)

Phase Six

Outro and References

Feature Improvements

As You May have already seen, the Project is still a prototype and we hope it can get more durable and easer to use and more effective in everyday use, VRB is intended to serv as easy to install/remove mounting, we hope we can achieve that in the feature with further development and higher technologies become available, lighter, faster, more of efficiency, and more affordable are our main goals to achieve the revolutionary ease of use.

Conclusion

In conclusion, this graduation project represents a significant milestone in our academic journey. It has been a challenging yet rewarding experience that has allowed us to apply the knowledge and skills acquired throughout our studies. Through this project, we have not only deepened our understanding of IT but also developed critical thinking, problem-solving, and research skills.

We are immensely grateful for the support and guidance provided by our supervisor, Dr. Mohanad Almoshriqy, whose expertise and encouragement have been invaluable. Additionally, we extend our heartfelt thanks to our parents, siblings, and friends, whose unwavering support has been a constant source of motivation.

As we look back on this journey, we are proud of the progress we have made and the obstacles we have overcome. This project has not only been an academic endeavor but also a personal growth experience. We are excited to carry forward the lessons learned and the skills honed into our future endeavors.

Thank you to everyone who has been part of this journey. Your contributions have made this achievement possible.

REFERENCES

[1] fi.edu HISTORY OF VIRTUAL REALITY. Harrod Lebron 2021

[2] history-computer.com HISTORY OF OCULUS. James Tomson 2019

[3] Headsets Images Each From the official website of the company.